Data Sheet

EXPERIMENT NO. 1

Title: - Study of Ripple Factor and Voltage Regulation Characteristics of Half-Wave Rectifier With Capacitor Filter (using Tinkercad software circuit).

Apparatus Required: -

Sl. No.	Apparatus / Component	Specification	Quantity

Important Formulae: -

Peak to peak ripple voltage Vr = $\frac{Vm}{fRC}$

Average DC output voltage with capacitive filter Vdc = Vm - $\frac{Vm}{2fRC}$

Ripple factor with capacitive filter $\gamma = \frac{1}{2\sqrt{3}fRC}$

Observation Table: -

With Capacitor Filter: -

C(μF)	R_L(KΩ)	V_{dc} (V)	V_{ac} (V)	R.F.(γ) (Practical) (V_{ac}/V_{dc})	R.F.(γ) (Theoretical) ($1/ 2\sqrt{3}fCR_L$)

Voltage Regulation: -

Condition	Output Voltage (V)
No Load (VNL)	
Full Load (VFL)	

Calculation: -

Voltage Regulation = [(VNL − VFL)/VFL] × 100 = _____ %

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Teacher’s Sign and Date